

A Predictive Machine Learning Framework for Ergonomic Risk in Last-Mile Quick-Commerce Delivery Operations

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In my capacity as supervisor of the above-mentioned work, I certify that the work presented in this Report is carried out under my supervision, and is worthy of consideration for the requirements of internship under IIITDM-SIES programme during the period May 2026 to July 2026.

Mentor's Name: Dr. Arunachalam Muthiah

Mentor's Signature

Abstract

Last-mile quick-commerce delivery riders (Blinkit, Zepto) face prolonged awkward postures, high delivery rates, heavy carrying loads and vehicle vibration, leading to musculoskeletal disorders (MSDs). Risk is not uniform across the workforce: age, tenure, daily hours, vehicle type, and carrying mode all shape the exposure profile, but this variation stays invisible to the platform. Standard ergonomic methods each cover part of the picture (RULA and QEC on the observation side, NMQ, NASA-TLX and Borg CR10 on self-report), and no single tool combines the six standard ergonomic risk factors into an automated per-rider screening. This work builds that screening on a cross-sectional sample of 182 riders from the Chengalpattu and Chennai regions collected in March and April 2026.

The pipeline runs in two stages. Stage 1 applies standard ergonomic methods (Borg CR10 action levels, RULA Table C thresholds, and sample-based cuts) to compute Low, Medium, or High risk labels for each factor from a 182-rider survey combined with 182 RULA and QEC observation records. Stage 2 trains supervised classifiers per factor using seven candidate algorithms (Logistic Regression, Decision Tree, Random Forest, Extra Trees, Hist Gradient Boosting, XGBoost, and Stacking) tuned via GridSearchCV with SMOTE inside 5-fold stratified cross-validation.

The five survey-derived factors achieved 58 to 62 percent cross-validation accuracy with macro AUC between 71 and 76 percent. The Posture model, which uses the 11 RULA components and 8 QEC scores in addition to the survey features, reached 97 percent accuracy and 98 percent AUC. Statistical analysis (chi-square and multivariable logistic regression) identified age, job duration, workload score, and fatigue score as the strongest individual predictors of self-reported discomfort. The final deliverable is a Streamlit web application that accepts the full 36-item questionnaire plus RULA and QEC observations and returns the six colour-coded risk levels along with per-factor recommendations.

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Abbreviations

MSD	Musculoskeletal Disorder
NMQ	Nordic Musculoskeletal Questionnaire
RULA	Rapid Upper Limb Assessment
QEC	Quick Exposure Check
NASA-TLX	NASA Task Load Index
CV	Cross-Validation
SMOTE	Synthetic Minority Over-sampling Technique
ML	Machine Learning
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
OR	Odds Ratio
CI	Confidence Interval
HistGBM	Histogram-based Gradient Boosting Machine
XGBoost	eXtreme Gradient Boosting
CART	Classification and Regression Trees
dph	Deliveries Per Hour
MSK	Musculoskeletal
CR10	Category-Ratio 10-point scale (Borg)
RF	Random Forest
ET	Extra Trees
LR	Logistic Regression
DT	Decision Tree
GBM	Gradient Boosting Machine
OvR	One-vs-Rest
SD	Standard Deviation

Symbols

n	sample size (n = 182 riders)
k	number of nearest minority neighbours used by SMOTE
λ	uniform random weight for SMOTE interpolation
β	regression coefficient in logistic regression
χ^2	chi-square test statistic
p	p-value (statistical significance)
dph	deliveries per hour, <code>deliveries_num / work_hours_num</code>
α	significance threshold ($\alpha = 0.05$)
df	degrees of freedom (chi-square test)
$F1$	harmonic mean of precision and recall

Chapter 1. Introduction

1.1 Background

Last-mile quick-commerce delivery riders for platforms like Blinkit and Zepto spend long hours on bikes, climb stairs, carry packages, and ride in heavy traffic. Prior Indian last-mile MSK studies [1, 2] report similar exposure patterns among app-based delivery riders. Over time this kind of work causes musculoskeletal disorders (MSDs): back pain, shoulder pain, wrist injuries, knee problems. The risk is not the same for every rider. Someone who is 22 and works 4 hours a day on a scooter is in a very different situation from a 45-year-old who works 10 hours a day on a motorbike with a handheld bag.

Standard ergonomic methods cover different facets of this problem. RULA and QEC handle the observational side (posture, body-region exposure), while NMQ, NASA-TLX, and Borg CR10 cover what the rider reports about pain, mental load, and perceived exertion. None of them on its own gives a complete picture. The project pulls all of them together into one per-rider profile across six ergonomic dimensions: Force, Repetition, Posture, Duration, Contact Stress, Vibration.

1.2 Motivation

The delivery-platform economy in India has grown rapidly in recent years. Riders form the largest occupational group in this economy and face physical demands that are qualitatively different from traditional warehouse or manufacturing work: long distances on two-wheelers, unstable load positioning, unpredictable traffic, weather exposure, and constant algorithmic time pressure. There is no automated tool that a platform can use to screen individual riders across all six standard ergonomic risk factors from a single questionnaire.

A per-rider risk profile would let a platform intervene early: shorten shifts for high-risk profiles, provide equipment reviews, offer targeted physiotherapy, and re-screen riders periodically. Without such a tool, risk stays invisible until MSDs surface as absenteeism or medical consultations. This project addresses that gap by combining published ergonomic assessment methods with modern classifiers into a single screening pipeline and web application.

1.3 Objectives of the work

1. Design a two-stage pipeline that combines a survey of quick-commerce delivery riders with standard ergonomic observation tools (RULA and QEC) into per-rider labels for six risk factors: Force, Repetition, Posture, Duration, Contact Stress, and Vibration.

2. Implement the pipeline reproducibly in Python (Jupyter notebooks, scikit-learn, XGBoost, imbalanced-learn, statsmodels) with a fixed random seed and end-to-end regeneration of every intermediate artefact.
3. Train and evaluate seven candidate ML algorithms per risk factor using 5-fold stratified cross-validation with SMOTE oversampling inside every fold, and report accuracy, macro F1, and macro AUC.
4. Identify and correct methodological issues discovered during modelling (class boundary degeneracy in the Repetition binning, feature leakage in the Duration model) and document each fix openly.
5. Deploy the trained models behind a Streamlit web application that accepts the full 36-item questionnaire plus the 11 RULA components and 8 QEC scores, and returns the six colour-coded risk levels with per-factor recommendations.

Chapter 2. Methodology

2.1 Two-stage design

The pipeline runs in two stages. Stage 1 assigns Low, Medium, or High labels for each of the six ergonomic risk factors using deterministic rules from standard methods (Borg CR10 action levels for Force, RULA Table C for Posture, sample terciles or fixed cuts for the rest); these labels are the ground truth Stage 2 has to learn. Stage 2 trains a supervised classifier per factor with per-target feature exclusions removing the inputs that define the label, and evaluates via 5-fold stratified cross-validation with SMOTE oversampling inside every training fold.

The two-stage split lets an ergonomist audit Stage 1 by hand and an ML reviewer audit Stage 2 independently, and the trained classifiers can then be deployed behind a web form that runs on a new rider's profile without re-applying the RULA and QEC worksheets.

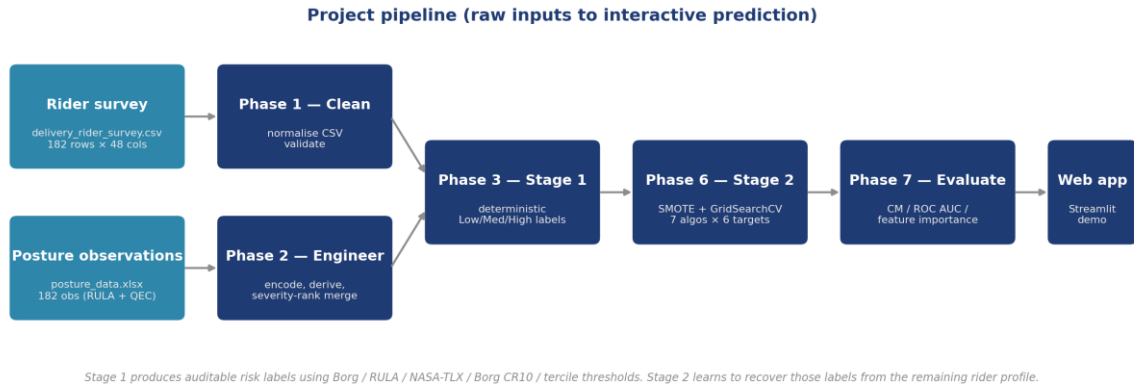


Figure 2.1: Pipeline overview from raw inputs to interactive prediction.

2.2 Study design and data sources

This is a cross-sectional study of Blinkit and Zepto quick-commerce delivery partners conducted in the Chengalpattu and Chennai regions in March-April 2026 using convenience sampling. The same 182 participants supplied both data streams: rider self-report through a Google Form and posture recorded during a delivery task by a trained observer. Eligibility was applied at the person level before either instrument.

Inclusion criteria

- Full-time delivery partners working on either the Blinkit or the Zepto platform.

- Minimum three months of continuous delivery experience.
- Using a two-wheeler (scooter or motorbike) for delivery.
- Both men and women, aged between 18 and 55 years.

Exclusion criteria

- Chronic musculoskeletal disorders that developed before the start of delivery work.
- Accidental surgery within the past three months.

2.2.1 Rider survey

delivery_rider_survey.csv holds 182 self-administered responses to a 36-item questionnaire covering demographics, work pattern, NMQ 12-month pain (9 body areas), 7-day discomfort (4 areas), five discomfort follow-ups, six NASA-TLX items (0-100), and six Borg CR10 items (0-10). After cleaning: 182 rows, 48 columns.

2.2.2 Posture observations

posture_data.xlsx holds 182 observation records, each carrying the 11 RULA components, 3 derived RULA Table scores, and 8 QEC scores. The rows lack rider identifiers; pairing to the survey uses the severity-rank merge in section 2.3.

2.3 Data cleaning and feature engineering

Phase 1 (01_data_cleaning.ipynb) normalises the raw CSV, standardises Yes/No answers, and validates row counts. Phase 2 (02_feature_engineering.ipynb) encodes ordered categorical bins to integers, assigns numeric midpoints to bin ranges, ranks vehicle type and carrying mode, aliases the 18 Yes/No items, and derives four composite features -- workload_score (mean of the six NASA-TLX items with satisfaction reversed), fatigue_score (mean of the six Borg CR10 items), force_exertion (the Borg lifting item), and vibration_index (vehicle_rank x work_hours_num) -- plus five product-form interactions so tree-based models can pick up combined effects.

The posture observations are merged into the survey via a severity-rank merge: riders are ranked by an exposure_severity index (normalised NMQ pain count + fatigue score + hours), posture rows by RULA Table C, and the two ranked lists are joined one-to-one. Caveats are noted in Chapter 5.

2.4 Stage 1: Deterministic risk scoring

Phase 3 (03_risk_scoring.ipynb) applies the following rules and writes risk_profile.csv:

- Force: Borg CR10 lifting (0-3 Low, 4-6 Medium, 7-10 High).
- Repetition: deliveries per hour, fixed cuts at 2.5 and 3.75.
- Duration: continuous riding hours (≤ 5 Low, 6-7 Medium, > 7 High).
- Vibration: vibration_index binned by sample tercile.
- Contact Stress: carrying_contact_rank x work_hours_num x age multiplier, sample tercile.

- Posture: RULA Table C (1-2 Low, 3-4 Medium, ≥ 5 High); training minimum is 3, so Posture has no Low class.

2.5 Statistical analysis

Phase 5 (05_stats.ipynb) runs two analyses on discomfort (binary: any NMQ 12-month pain). A chi-square test of independence (scipy.stats.chi2_contingency) checks whether each risk factor is associated with discomfort at p less than 0.05. Second, a multivariable logistic regression (statsmodels.api.Logit) estimates the effect of eleven predictors while controlling for the others; results are reported as odds ratios with 95 percent confidence intervals.

2.6 Stage 2: Machine learning pipeline

Phase 6 (06_modeling.ipynb) trains a classifier per target from seven candidate algorithms. Each sits inside an imblearn.Pipeline with SMOTE upstream so resampling only touches training folds. GridSearchCV tunes hyperparameters on macro-F1 inside 5-fold stratified CV, and the best model per target is refit on the full sample and saved as outputs/models/best_<factor>.pkl.

2.6.1 Logistic Regression

Regularised linear classifier used as the baseline (L2 penalty, class_weight balanced). Any tree-based method that fails to beat it on a factor is either overfitting or getting nothing from feature interactions.

2.6.2 Decision Tree

Single CART tree with max_depth in {3, 5, 7} and min_samples_leaf in {1, 3, 5} tuned via grid search. Included as the ensembling floor; not competitive on its own for any of the six targets.

2.6.3 Random Forest

Bagged ensemble of 300-500 trees with bootstrap sampling and $\sqrt{p}$ features per split. Low-variance predictions and mixed feature-type handling made it the winner on Repetition, Duration, and Contact Stress.

2.6.4 Extra Trees

Like Random Forest but with fully random splits per node. The extra randomisation lowers between-tree correlation and helped it win on Vibration.

2.6.5 Histogram Gradient Boosting

Sequential residual-fit ensemble with histogram-binned features for training speed. Won on Force (Borg CR10 cutoffs are nonlinear) and Posture (strongest signal from RULA + QEC features).

2.6.6 XGBoost

Gradient-boosted trees with L2 regularisation on the leaf weights. Placed second on several targets but did not win any at the default search budget.

2.6.7 Stacking Classifier

Meta-learner taking the out-of-fold predictions of Random Forest, Extra Trees, HistGBM, and XGBoost as inputs and fitting a Logistic Regression on top. Failed to beat the strongest base learner because the four base learners largely agreed.

2.6.8 SMOTE and cross-validation

SMOTE synthesises new minority-class samples along segments between k nearest minority neighbours. Wrapping it in `imblearn.Pipeline` keeps the synthesis inside training folds only, so validation-fold accuracy is never inflated. `k_neighbors` is set per target below the minority training-fold count; `GridSearchCV` picks hyperparameters on macro-F1 with `StratifiedKFold(5, shuffle=True, random_state=42)`.

2.6.9 Per-target feature exclusions

Each target excludes the survey inputs that directly define its Stage-1 label (otherwise the model would memorise the rule). Posture additionally receives the 11 RULA components and 8 QEC scores, giving 63 total features.

Target	Excluded features	Final feature count
Force	force_exertion, force_x_age	42
Repetition	deliveries_num, work_hours_num, deliv_x_days	41
Duration	work_hours_num, vibration_index	42
Vibration	vibration_index, vehicle_rank, work_hours_num	41
Contact Stress	carrying_contact_rank, work_hours_num	42
Posture	(none from survey pool)	$44 + 11 + 8 = 63$

Table 2.1: Per-target feature exclusions to prevent label leakage.

2.7 Evaluation

Phase 7 (`07_evaluation.ipynb`) uses `cross_val_predict` for out-of-fold predictions and writes per-class precision / recall / F1 / support, confusion matrices, one-vs-rest ROC curves with per-class AUC (macro AUC is the unweighted mean), and feature importance (`feature_importances_` for tree models, `abs(coef_)` for Logistic Regression).

Chapter 3. Work Done

3.1 *Notebook pipeline*

The full pipeline is implemented across seven Jupyter notebooks (01 through 07) plus a Streamlit application. Every notebook is designed to run standalone from its previous input file, so the whole pipeline can be regenerated end to end with a fixed random seed of 42.

- 01_data_cleaning normalises the raw CSV and validates row counts.
- 02_feature_engineering encodes categoricals, derives composite scores, aliases the 18 binary items, and merges the posture observations by severity rank.
- 03_risk_scoring applies standard thresholds to compute Low/Medium/High labels per risk factor.
- 04_eda produces demographics, NMQ prevalence, risk distribution, discomfort-by-group, and correlation heatmap figures.
- 05_stats runs chi-square and multivariable logistic regression.
- 06_modeling runs SMOTE plus GridSearchCV across 7 algorithms per target and keeps the best by F1 macro.
- 07_evaluation produces confusion matrices, ROC curves, classification reports, and feature importance plots.

3.2 *Instrument design and preparation*

Two data-collection instruments were prepared before fieldwork began. The self-report instrument was a 36-item Google Form structured in five modules: demographics and work pattern (seventeen items), the Nordic Musculoskeletal Questionnaire covering 12-month pain across nine body areas (Q18) and 7-day discomfort across four areas (Q19), five discomfort-related follow-up items on performance impact, leave taken, medical consultation, and worsening triggers (Q20 to Q24), six NASA-TLX workload items on 0 to 100 sliders (Q25 to Q30), and six Borg CR10 perceived-exertion items on 0 to 10 sliders (Q31 to Q36). The NASA-TLX performance item is worded as satisfaction (higher = better performance), so at analysis time it is reversed to 100 minus the raw value; the workload composite is then the unweighted mean of the six items (Raw TLX).

The observation instrument was a paper form carrying the eleven standard RULA elements (upper arm, lower arm, wrist, wrist twist, neck position, trunk position, legs, muscle A, force A, muscle B, force B), the three RULA Table lookup scores (A, B, C), and the eight QEC region and exposure scores (back, shoulder / arm, wrist / hand, neck, driving, vibration, work pace, stress). Both instruments were piloted with three delivery partners outside the study region so the item wording, response options, and observer worksheet layout could be revised before real fieldwork began.

3.3 Data collection execution

Fieldwork ran across the Chengalpattu and Chennai regions between March and April 2026, following the eligibility criteria and instruments described in Section 2.2. Recruitment used convenience sampling at platform handover points where delivery partners regularly cluster between shifts.

Each participant gave verbal consent before either instrument was administered. The 36-item Google Form was completed on a researcher-provided phone in about 15 to 20 minutes. Posture was recorded on paper by a trained observer during a natural delivery task, capturing the 11 RULA elements and 8 QEC region / exposure scores in one pass. Riders whose demographic or NMQ responses were incomplete after follow-up were excluded and are not part of the final 182. Data was anonymised at the point of collection; no rider identifiers appear in the released `model_ready.csv` or `posture_data.xlsx` files.

3.4 Data cleaning, encoding, and integration

Data preparation ran in two notebooks. `01_data_cleaning` loaded the raw Google Form CSV, standardised text answers to canonical Yes / No values, corrected a handful of column-name typos, and dropped rows that were incomplete after follow-up. The cleaned CSV was 182 rows and 48 columns.

`02_feature_engineering` converted the cleaned CSV into the model-ready feature set. Ordered categorical bands (age, education, income, job duration, working hours, working days, deliveries per day, rest break) were encoded to consecutive integers preserving order. Vehicle type and carrying mode were converted to ranks so motorbike ranks higher than scooter for vibration exposure and handheld ranks higher than backpack and bike storage for contact stress. Composite scores were computed: `workload_score` as the mean of the six NASA-TLX items (with the dissatisfied item reversed), `fatigue_score` as the mean of the six Borg items, `force_exertion` as the Borg lifting item, and `vibration_index` as `vehicle_rank` times `work_hours_num`. Five product-form interactions were added (workload times fatigue, workload times age, force times age, fatigue times job duration, deliveries times days). The eighteen NMQ and 7-day items and the five outcome follow-ups were converted to short-named binary aliases (`nmq_neck`, `d7_lower_back`, `out_reduced_perf`, and so on).

The posture `xlsx` did not carry rider identifiers, so it had to be paired with the survey without a natural join key. A severity-rank merge was used: riders were ranked by an aggregate discomfort score (NMQ pain count plus fatigue score plus daily hours), observations were ranked by RULA Table C, and the two ranked lists were matched one-to-one by position. The merged `model_ready.csv` contains 182 rows and 121 columns.

3.5 Statistical analysis and model training runs

Two independent statistical analyses were run in 05_stats on the model-ready CSV. First, six chi-square tests of independence checked whether each Stage-1 risk-factor band was associated with a binary discomfort outcome; the tests reported chi2, degrees of freedom, and a Bonferroni-adjusted p-value. Second, a multivariable logistic regression fitted with statsmodels estimated odds ratios for eleven predictors (age, income, education, job duration, workload score, fatigue score, force exertion, vibration index, contact rank, hours, days) with self-reported discomfort as the outcome. Results were reported as OR with 95 percent confidence intervals.

Model training ran in 06_modeling. Every one of the seven candidate algorithms was wrapped in an imblearn.Pipeline of SMOTE followed by the classifier and hyperparameter-tuned via GridSearchCV. For each of the six risk-factor targets, this expanded to a full five-fold stratified cross-validation over the algorithm's hyperparameter grid. Across the six targets and seven algorithms, forty-two target-algorithm training runs were carried out. Best-by-macro-F1 was selected per target and the winning pipeline was refit on the full sample and persisted with joblib.dump to outputs/models/best_<factor>.pkl. The per-target feature exclusions listed in Section 2.6.9 were applied before each training run so no model could memorise its Stage-1 rule.

07_evaluation produced the reporting artefacts. cross_val_predict was used to generate out-of-fold predictions on the training set, and the following outputs were written: per-class precision, recall, F1, and support numbers via classification_report; confusion matrices per target; one-vs-rest ROC curves with per-class AUC (macro AUC is the unweighted mean of per-class AUCs); and feature importance plots (feature_importances_ for the tree-based winners, absolute value of coef_ for the Logistic Regression floor).

3.6 Web application development and deployment

The Streamlit web application (app/streamlit_app.py) exposes a six-section form that mirrors the entire 36-item questionnaire plus the 11 RULA components and 8 QEC scores. Selectboxes cover demographics and work pattern; radios cover NMQ, 7-day, and outcome items; sliders cover NASA-TLX (0 to 100) and Borg CR10 (0 to 10); selectboxes cover RULA components; and number inputs cover the QEC scores with min / max ranges matched to the observed training data. Three sample-profile buttons (Low, Average, and High risk) fill the entire form on click so the reviewer can trigger a prediction in one step.

On form submission, encode_rider assembles the 63-feature input vector, and predict_risks runs each of the six saved models. The Results page renders a coloured metric strip (green for Low, amber for Medium, red for High), a horizontal Altair bar chart of the same six factors sorted to match the metric strip, a summary banner that flags high-burden profiles, per-factor recommendation cards, and an

expandable JSON view of the exact 63 feature values that were fed to the models. The app also has a Methodology page that summarises the pipeline and an About page carrying the institutional context.

The app was deployed to Streamlit Community Cloud from the public GitHub repository. Deployment required pinning every runtime dependency in requirements.txt to the exact versions the models were trained with (streamlit 1.58, pandas 2.3, scikit-learn 1.7, xgboost 3.2, imbalanced-learn 0.14, plotly 6.8), because sklearn and xgboost pickles are not portable across minor versions. runtime.txt pins Python 3.13 to match the training environment. A .streamlit/config.toml carries the dark theme and a client.toolbarMode = viewer flag; CSS in the app suppresses the Fork button and 'View source on GitHub' icon that Streamlit Community Cloud injects on public-repo deploys. Every git push to the main branch auto-rebuilds and redeploys the app in about 30 seconds.

3.7 Iterations and fixes

3.7.1 Repetition binning correction

The initial Phase 3 binning used pandas.qcut(q=3) on deliveries_per_hour. The 66.7th percentile fell exactly on 3.889 dph (equal to 35 deliveries over 9 hours, the worst real combination in the sample). Fifty-five of the 182 riders tied at that value, and qcut is right-inclusive at the upper edge, so all 55 fell into Medium. This left the Stage-2 model unable to predict High for the worst real schedule. The fix replaces qcut with fixed cuts at 2.5 and 3.75. The Stage-1 High count grew from 19 to 74. Cross-validation accuracy dropped from 74 percent to 62 percent because the model now solves a real 3-class problem instead of memorising a 19-row minority.

3.7.2 Duration leakage correction

An earlier Phase 6 run reported 100 percent cross-validation accuracy on Duration. Investigation showed the trees were reconstructing work_hours_num from vibration_index, which equals vehicle_rank multiplied by work_hours_num. Adding vibration_index to the Duration exclusion list and capping max_depth returned Duration to a realistic 61 percent accuracy with 76 percent macro AUC.

3.7.3 Posture model with RULA and QEC inputs

The Posture target was initially trained only on the survey feature pool, which meant it could only infer posture indirectly from pain, fatigue, and hours. Adding the 11 RULA components and 8 QEC scores from the observation.xlsx as explicit features moved Posture accuracy from 89 to 97 percent and the overfit gap from 0.088 to 0.028. Per the label-leakage rule, the three derived RULA Table scores were excluded because posture_risk is rula_table_c binned.

Chapter 4. Results and Discussions

4.1 Sample profile

The sample of 182 riders is 152 male and 30 female. Age skews young: 78 riders are under 25, 66 in the 25-35 band, 30 in 36-45, and 8 in the 46-and-above band. Roughly half the sample (49 percent) works more than 8 hours per day. Ninety-four ride scooters and 88 ride motorbikes. The platform split is 97 Blinkit, 80 Zepto, and 5 who work both platforms.

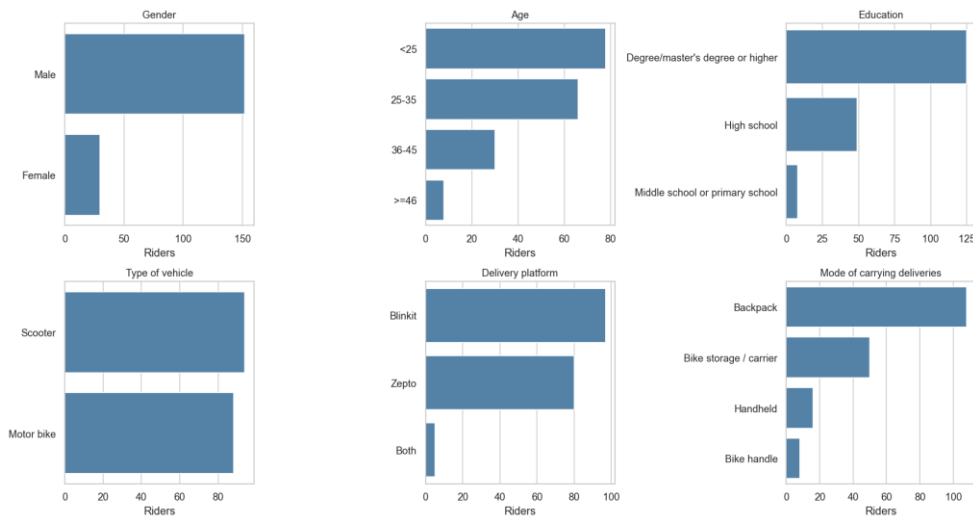


Figure 4.1: Sample profile: age, platform, vehicle, and carrying mode.

4.2 NMQ pain prevalence and statistical predictors

84.6 percent of the riders reported pain in at least one body area in the last 12 months. Table 4.1 lists the prevalence per body area, sorted from highest to lowest.

Body area	Prevalence
Lower back	61.5%
Upper back	49.5%
Shoulders	46.7%
Wrists / Hands	45.1%
Hips / Thighs	41.2%
Ankles / Feet	40.7%
Knees	39.6%
Neck	37.9%
Elbows	33.5%

Table 4.1: NMQ 12-month pain prevalence per body area.

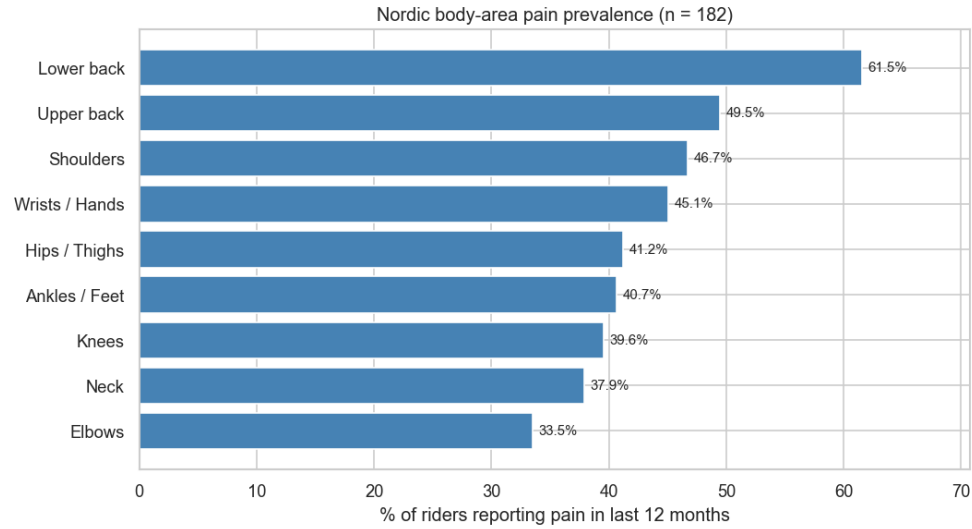


Figure 4.2: NMQ 12-month pain prevalence per body area.

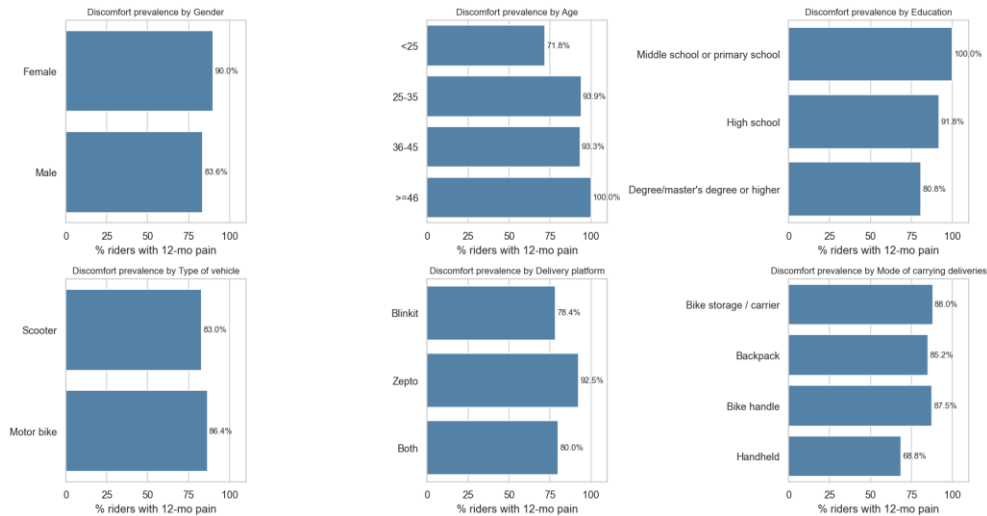


Figure 4.3: Discomfort prevalence broken down by demographic group.

The chi-square test of independence identifies Posture, Repetition, and Force as the three factors with a significant association with self-reported discomfort at p less than 0.05:

Factor	chi-square	p	Significant
Posture	45.665	<0.001	Yes
Repetition	8.617	0.014	Yes
Force	6.718	0.035	Yes
Duration	0.623	0.733	No
Vibration	0.600	0.741	No
Contact Stress	0.544	0.762	No

Table 4.2: Chi-square test: risk factor vs self-reported discomfort.

The multivariable logistic regression identifies seven significant individual predictors of discomfort while controlling for the others. Age is the strongest: each step up in the age band multiplies the odds of discomfort by 3.58 (95 percent CI 1.70 to 7.56, $p = 0.0008$).

Predictor	OR	95% CI	p
Workload score (per unit)	1.06	1.02 to 1.09	0.0005
Age (per band)	3.58	1.70 to 7.56	0.0008
Job duration (per band)	2.89	1.54 to 5.41	0.001
Fatigue score (per unit)	1.43	1.13 to 1.80	0.003
Income (per band)	2.00	1.25 to 3.20	0.004
Education (per band)	0.33	0.12 to 0.94	0.04
Deliveries per day	1.05	1.00 to 1.10	0.045

Table 4.3: Significant predictors of discomfort from multivariable logistic regression.

4.3 Stage-1 risk distribution

Factor	Low	Medium	High
Force	90	57	35
Repetition	26	82	74
Duration	37	56	89
Vibration	67	68	47
Contact Stress	68	58	56
Posture	0	29	153

Table 4.4: Stage-1 risk band counts per factor ($n = 182$).

Posture, Duration, and Repetition are the three factors where the High band dominates. Posture is at 84 percent High (153 of 182 observations); Duration is at 49 percent High; Repetition is at 41 percent High after the binning fix.

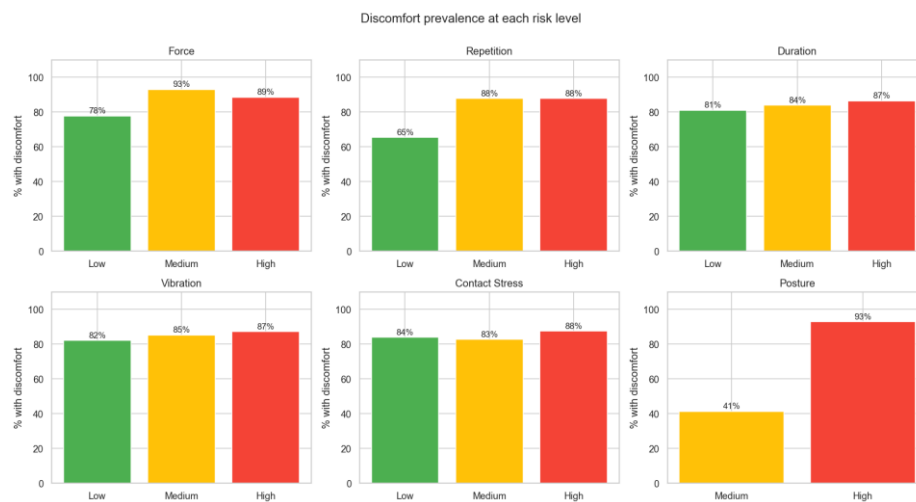


Figure 4.4: Discomfort prevalence within each Low / Medium / High band, per risk factor.

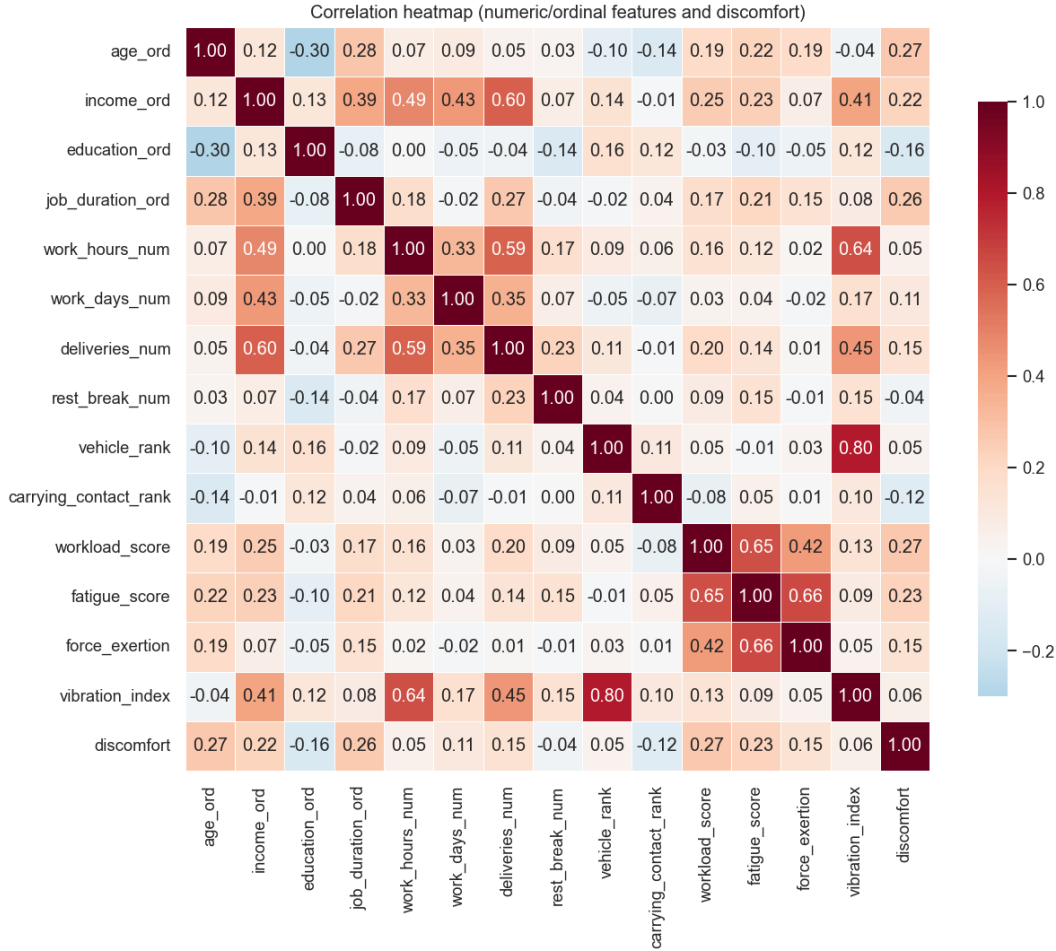


Figure 4.5: Pearson correlation matrix across the numeric feature pool, upper triangle masked.

4.4 Stage-2 model performance

Factor	Best model	Acc.	F1 macro	Macro AUC	Features
Force	HistGradientBoosting	62%	57%	71%	42
Repetition	RandomForest	62%	57%	73%	41
Duration	RandomForest	61%	58%	76%	42
Vibration	ExtraTrees	58%	57%	72%	41
Contact Stress	RandomForest	60%	59%	74%	42
Posture	HistGradientBoosting	97%	95%	98%	63

Table 4.5: Best Stage-2 model per risk factor after 5-fold stratified CV.

The five survey-derived factors land between 58 and 62 percent cross-validation accuracy with macro AUC in the 71 to 76 percent range. Posture reaches 97 percent accuracy and 98 percent macro AUC because it is the only model that receives real observation inputs (the 11 RULA components and 8 QEC scores) on top of the survey features.

4.5 Per-class metrics and feature importance

Factor	Low	Medium	High
Force	0.798	0.580	0.760
Repetition	0.745	0.669	0.782
Duration	0.822	0.649	0.797
Vibration	0.775	0.615	0.779
Contact Stress	0.763	0.651	0.791
Posture	-	-	0.984

Table 4.6: Per-class ROC AUC (one-vs-rest) for the best model per factor.

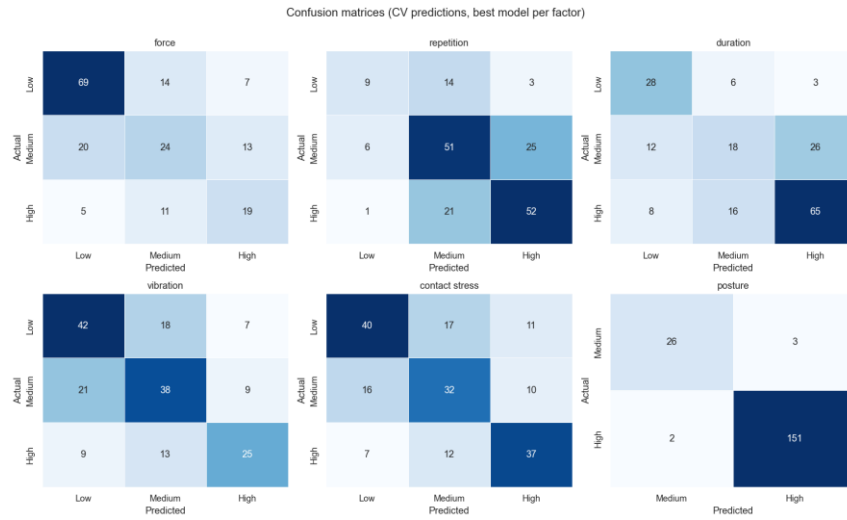


Figure 4.6: Confusion matrices for the best model per factor (rows = true class, columns = predicted class).

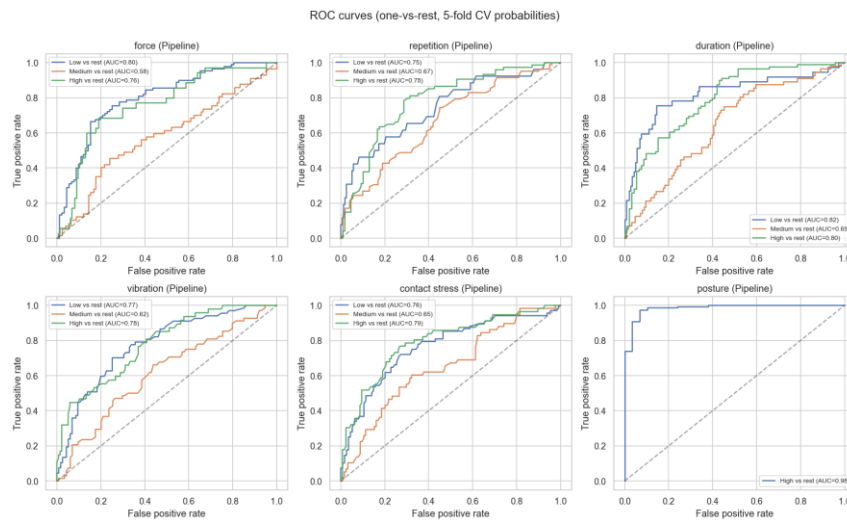


Figure 4.7: ROC curves (one-vs-rest) for the best model per factor.

Table 4.7 gives per-class precision, recall, F1, and support for the best model per factor. The Repetition High class has 65 percent precision and 70 percent recall; the Duration High class has 69 percent precision and 73 percent recall.

Factor	Class	Precision	Recall	F1	Support
Force	Low	0.734	0.767	0.750	90
Force	Medium	0.490	0.421	0.453	57
Force	High	0.487	0.543	0.514	35
Repetition	Low	0.562	0.346	0.429	26
Repetition	Medium	0.593	0.622	0.607	82
Repetition	High	0.650	0.703	0.675	74
Duration	Low	0.583	0.757	0.659	37
Duration	Medium	0.450	0.321	0.375	56
Duration	High	0.691	0.730	0.710	89
Vibration	Low	0.583	0.627	0.604	67
Vibration	Medium	0.551	0.559	0.555	68
Vibration	High	0.610	0.532	0.568	47
Contact Stress	Low	0.635	0.588	0.611	68
Contact Stress	Medium	0.526	0.552	0.539	58
Contact Stress	High	0.638	0.661	0.649	56
Posture	Medium	0.929	0.897	0.912	29
Posture	High	0.981	0.987	0.984	153

Table 4.7: Per-class precision, recall, F1, and support.

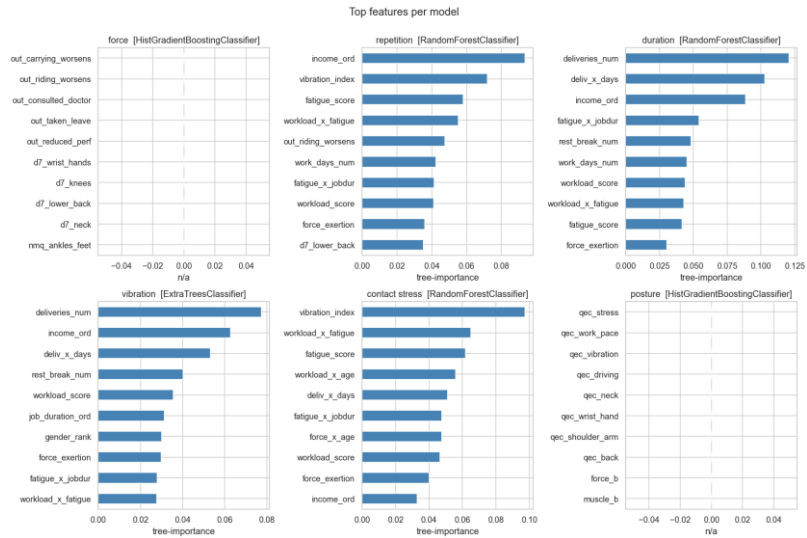


Figure 4.8: Top 10 features by importance for the best model per factor.

Factor	Best model	Hyperparameters	SMOTE k
Force	HistGBM	max_depth=None, learning_rate=0.05	5
Repetition	RandomForest	n_estimators=300, max_depth=None	5
Duration	RandomForest	n_estimators=500, max_depth=5	5
Vibration	ExtraTrees	n_estimators=300, max_depth=None	5
Contact Stress	RandomForest	n_estimators=500, max_depth=None	5
Posture	HistGBM	max_depth=5, learning_rate=0.05	5

Table 4.8: Winning hyperparameters per target after GridSearchCV.

4.6 Web application demonstration

Figures 4.9 through 4.13 show one full pass through the form and the resulting prediction output. Three sample-profile shortcuts at the top of the form (Low, Average, High risk) pre-fill every field for a one-click demo.

The screenshot displays the 'Rider Assessment' web application. On the left is a dark sidebar with navigation links: Home, Assessment (selected), Results, Methodology, and About. Below these links is the title 'Ergonomic Risk Screening Tool' and the text 'BITOM-QES | SIES Research | Deliverable'. At the bottom of the sidebar, it lists 'A/LA VISHNU VAREHAN' and 'Mentor: Dr. Arunachalam Muthiah'.

The main content area has a title 'Rider Assessment' and a subtitle: 'Answer the 36-item questionnaire below plus the RULA and QEC observation scores. Or load a sample profile to autofill every field, then click Predict.' Below this are three sample profile shortcuts: 'Low-risk rider', 'Average rider', and 'High-risk rider'. A 'SECTIONS:' bar contains tabs for 'DEMOGRAPHIC', 'NMQ', 'NASA-TLX', 'BORG CR18', 'RULA', and 'QEC', with 'DEMOGRAPHIC' currently selected.

The 'Demographic' section contains 17 questions (Q1 to Q17) arranged in a grid. Each question has a dropdown menu for selection. The questions are:

- 1. Gender: Male
- 2. Age: 25-35
- 3. Education: Degree / Master's
- 4. Region: Local
- 5. Marital status: Unmarried
- 6. Delivery platform: Blinkit
- 7. Job duration (experience): 12-24 months
- 8. Monthly income (INR): 20,000-35,000
- 9. Type of vehicle: Motor bike
- 10. Mode of carrying deliveries: Backpack
- 11. Working hours per day: 6-8 hrs
- 12. Working days per week: 5-6
- 13. Number of deliveries per day: 21-30
- 14. Duration of rest break: 5-15 min
- 15. Type of break: Uneven
- 16. Tobacco consumption: Never
- 17. Alcohol consumption: Never

Figure 4.9: Web app: sample-profile shortcuts and demographic section (Q1 to Q17).

The screenshot shows the 'Nordic Musculoskeletal Questionnaire (NMQ)' section. It contains questions Q18 to Q24. Questions Q18 to Q20 are about pain, discomfort, or numbness in various body areas (Neck, Shoulders, Upper back, Lower back, Elbows, Wrists / Hands, Hips / Thighs, Knees, Ankles / Feet) over the past 12 months, with radio button options for 'No' and 'Yes'. Question Q19 asks about discomfort in the past 7 days for Neck, Lower back, Knees, and Wrist / Hands, also with radio button options. Questions Q21 to Q24 are follow-up questions about delivery performance, body pain, long hours, and heavy packages, with radio button options for 'No' and 'Yes'.

Below the NMQ section is the 'NASA-TLX workload (Low 0 - 100 High)' section. It contains four horizontal sliders:

- Q25: How mentally demanding was your delivery shift? (Scale 0 to 50)
- Q26: How physically demanding was your delivery shift? (Scale 0 to 50)
- Q27: How DISSATISFIED were you with your ability to complete deliveries accurately and on time? (Scale 0 to 100)
- Q28: How much effort did you put in to maintain performance? (Scale 0 to 100)

Figure 4.10: Web app: Nordic Musculoskeletal Questionnaire section (Q18 to Q24).

Figure 4.11: Web app: NASA-TLX and Borg CR10 sliders (Q25 to Q36).

Figure 4.12: Web app: RULA (11 items) and QEC (8 scores) observation sections above the Predict button.

Figure 4.13: Web app: predicted risk profile output with colour-coded level cards, radar chart, tabular result, and per-factor recommendations.

Chapter 5. Conclusions and Extensions

This project designed, implemented, and deployed a per-rider ergonomic risk screening pipeline for last-mile quick-commerce delivery workers, turning a 36-item self-report questionnaire and a RULA + QEC observation record into six standardised risk levels across Force, Repetition, Posture, Duration, Contact Stress, and Vibration. The five survey-derived factors land at 58-62 percent cross-validation accuracy with macro AUC between 71 and 76 percent; the Posture model, which receives the RULA and QEC observation inputs directly, reaches 97 percent accuracy and 98 percent AUC. Chi-square identifies Posture, Repetition, and Force as the risk factors significantly associated with self-reported discomfort, and multivariable logistic regression flags age (OR 3.58 per band), job duration (OR 2.89), workload score, and fatigue score as the strongest individual predictors.

The main limitations are the self-report nature of most input variables, the severity-rank merge used to pair the identifier-less posture xlsx with the survey, and the small sample ($n = 182$) that keeps ML variance visible. Two methodological corrections were made and openly documented: Repetition binning moved from `qcut` to fixed cuts to remove a boundary tie that hid the worst real schedule, and `vibration_index` was added to the Duration exclusion list to remove indirect leakage.

Five interventions for a delivery-platform operator follow directly from the results: (i) cap daily hours (49 percent of the sample exceeds 8 hours per day); (ii) invest in posture-focused equipment and training (84 percent of observed postures at RULA grand score 5 or above, corresponding to action level 3 or 4); (iii) encourage bike-storage-box carrying over handheld bags; (iv) age-targeted MSD screening for the 36-and-above cohort; and (v) platform-level workload management to reduce the NASA-TLX component of the workload score, a significant discomfort predictor at $p = 0.0005$.

Future extensions include a longitudinal follow-up so a causal claim can replace the cross-sectional association, per-rider RULA and QEC observations to eliminate the severity-rank merge, wearable accelerometer data as a Vibration proxy, and a periodic re-training schedule so the screening tool tracks a changing workforce. The two-stage design leaves room for each of these without disturbing the audited Stage-1 labels.

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